Solution: Count of Palindromic Substrings

Problem Statement

Given a string, find the total number of palindromic substrings in it. Please note we need to find the total number of substrings and not subsequences.

Example 1:

```
Input: "abdbca"
Output: 7
Explanation: Here are the palindromic substrings, "a", "b", "d", "b",
"c", "a", "bdb".
```

Example 2:

```
Input: = "cddpd"
Output: 7
Explanation: Here are the palindromic substrings, "c", "d", "d", "p",
"d", "dd", "dpd".
```

Example 3:

```
Input: = "pqr"
Output: 3
Explanation: Here are the palindromic substrings,"p", "q", "r".
```

Solution

This problem follows the `"Longest Palindromic Subsequence"` pattern, and can be easily converted to `"Longest Palindromic Substring"`. The only difference is that instead of calculating the longest palindromic substring, we will instead count all the palindromic substrings.

Let's jump directly to the bottom-up dynamic programming solution:

Java

```java
class Solution {

  public int findCPS(String st) {
    // dp[i][j] will be 'true' if the string from index 'i' to index 'j' is a
    // palindrome
    boolean[][] dp = new boolean[st.length()][st.length()];
    int count = 0;

    // every string with one character is a palindrome
    for (int i = 0; i < st.length(); i++) {
      dp[i][i] = true;
      count++;
    }

    for (int startIndex = st.length() - 1; startIndex >= 0; startIndex--) {
      for (int endIndex = startIndex + 1; endIndex < st.length(); endIndex++) {
        if (st.charAt(startIndex) == st.charAt(endIndex)) {
          // if it's a two character string or if the remaining string is a palindrome t
          if (endIndex - startIndex == 1 || dp[startIndex + 1][endIndex - 1]) {
            dp[startIndex][endIndex] = true;
```

Run Code Result | Stdout

You must run your code first

Execute

The time and space complexity of the above algorithm is $O(n^2)$, where 'n' is the length of the input string.

← Previous
Count Of Palindromic Substrings

Next →
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Mark as Completed